Computational Genomics – Exercise 2

Library Complexity

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1 Setup and notation

Supplementary material. Two short primers accompany this submission as background material I wrote while working through the unfamiliar parts of the exercise: `subsampling_primer.pdf` (the Heck rarefaction / Monte Carlo subsampling estimator used in Q2(a)) and `stats_primer.pdf` (the chi-squared goodness-of-fit, Wilks' dof correction, and the per-bin LRT used in Q2(b) and Q3(b)). They are cited from the relevant sections below; the main report is self-contained without them.

A note on notation. `Ex2.pdf` and the Lecture 10 notes use inconsistent symbols (notably, N and M are swapped). This report uses the **Lecture 10 convention** throughout; mapping:

Quantity	Ex2.pdf	This report (Lecture 10)
True # molecules in the library (complexity)	N	M
# sequenced reads	M	N
# unique molecules observed	M_u	U
Per-molecule read count	k_i	n_j
Rate per molecule	$\lambda = M/N$	$\lambda = N/M$

Model and identities. Each of the M original molecules has a unique sequence (deduplication recovers U exactly), and per-molecule read counts are i.i.d. $n_j \sim \text{Poisson}(\lambda)$ ($j = 1,\ldots,M$), with $N = \sum_j n_j$ and $U = \sum_j \mathbf{1}\{n_j > 0\}$. Intuitively, λ is the average number of reads each molecule receives – bigger λ means deeper effective sequencing per molecule. Following Ex2.pdf, λ is kept as a free parameter (needed in Q3). Immediate consequences: $\Pr(n_j > 0) = 1 - e^{-\lambda}$, $\mathbb{E}[U] = M(1 - e^{-\lambda})$ (the library-saturation curve, linear at small λ, capped at M as $\lambda \to \infty$; see Figure 1), $\mathbb{E}[N] = \lambda M$, and $\mathbb{E}[U]/\mathbb{E}[N] = (1 - e^{-\lambda})/\lambda$.

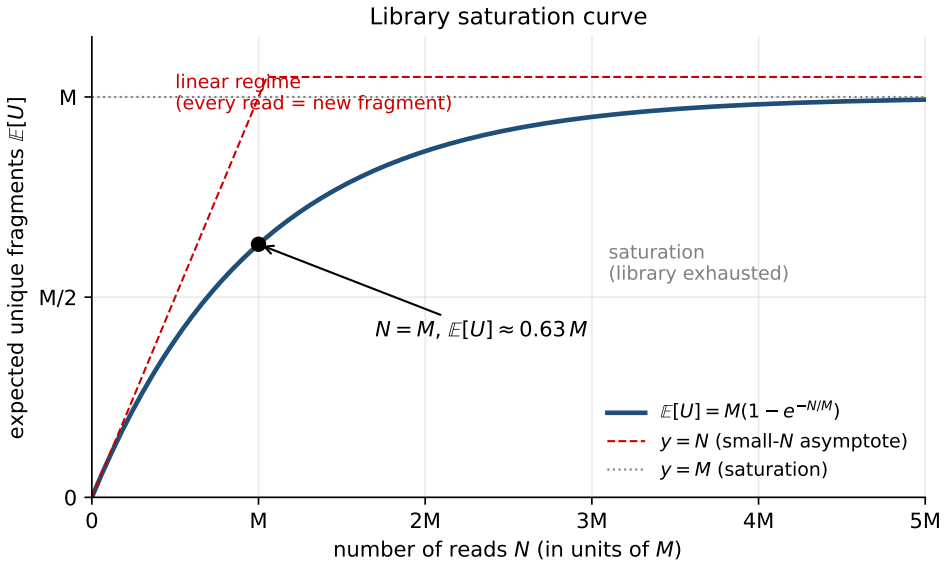


Figure 1: Saturation curve $\mathbb{E}[U] = M(1 - e^{-N/M})$. Linear for $N \ll M$, capped at M for $N \gg M$. At $N = M$, $\mathbb{E}[U] \approx 0.63\,M$.

Procedure from the Setup. Substituting observed for expected, then solving for $\hat{\lambda}$ and $\hat{M}$:

$$\frac{U}{N} \approx \frac{1 - e^{-\lambda}}{\lambda} \quad \Longrightarrow \quad \hat{M} \approx \frac{U}{1 - e^{-\hat{\lambda}}} = \frac{N}{\hat{\lambda}}.$$

$\hat{\lambda}$ is solved numerically from the first equation. The two forms of $\hat{M}$ on the right are equal by direct rearrangement of that equation: multiplying $\hat{\lambda}/(1 - e^{-\hat{\lambda}}) = N/U$ by $U/\hat{\lambda}$ gives $U/(1 - e^{-\hat{\lambda}}) = N/\hat{\lambda}$.

2 Question 1 – Maximum-likelihood estimator for λ

2.1 (a) The observation likelihood $P_{\text{obs}}(n_j; \lambda)$

Question (Ex2.pdf). What is $P_{\text{obs}}(k_i; \lambda)$? Hint: if $k_i = 0$ the molecule is not observed and we do not know it exists.

The model has $n_j \sim \text{Poisson}(\lambda)$, but the plain Poisson mass function $P(n_j; \lambda)$ does not suffice on its own: P_{obs} must condition on $n_j > 0$, since we do not observe molecules with $n_j = 0$ – they are invisible to us. Therefore:

$$\begin{aligned} L(\lambda) &= \prod_{j\,:\,n_j>0} P_{\text{obs}}(n_j; \lambda) = \prod_{j\,:\,n_j>0} P(n_j;\, \lambda,\, n_j > 0) \\ &\overset{\text{cond. prob.}}{=} \prod_{j\,:\,n_j>0} \frac{P_\lambda(n_j,\, n_j > 0)}{P_\lambda(n_j > 0)} \quad \overset{\{n_j=k\} \subseteq \{n_j>0\}}{=} \quad \prod_{j\,:\,n_j>0} \frac{P_\lambda(n_j)}{1 - P_\lambda(n_j = 0)} \\ &= \prod_{j\,:\,n_j>0} \frac{e^{-\lambda}\,\dfrac{\lambda^{n_j}}{n_j!}}{1 - e^{-\lambda}}. \end{aligned}$$

2.2 (b) Log-likelihood and sufficient statistics

Question (Ex2.pdf). Write the analytical form of the (log-)likelihood. What are the sufficient statistics?

From (a), $L(\lambda) = \prod_{j\,:\,n_j>0} \dfrac{e^{-\lambda}\,\lambda^{n_j}/n_j!}{1-e^{-\lambda}}$. Taking the log:

$$\begin{aligned}
\ell(\lambda) \;:=\; \log L(\lambda) &= \sum_{j\,:\,n_j>0} \log\left(\frac{e^{-\lambda}\,\dfrac{\lambda^{n_j}}{n_j!}}{1-e^{-\lambda}}\right) \\
&= \sum_{j\,:\,n_j>0} \Big[\log(e^{-\lambda}) + \log(\lambda^{n_j}) - \log(n_j!) - \log(1-e^{-\lambda})\Big] \\
&= \sum_{j\,:\,n_j>0} \Big[-\lambda + n_j\log(\lambda) - \log(n_j!) - \log(1-e^{-\lambda})\Big].
\end{aligned}$$

Since λ is the unknown parameter, the minimal sufficient statistic for λ is the pair (U, N) with $U = \#\{j : n_j > 0\}$ and $N = \sum_{j:n_j>0} n_j$. (The full vector $\{n_j : n_j > 0\}$ is trivially sufficient too, but the factorisation in (c) shows the meaningful compression.)

2.3 (c) Solving for $\hat\lambda = \arg\max_\lambda L(\lambda)$

Question (Ex2.pdf). How would you solve for $\hat\lambda = \arg\max_\lambda L(\lambda)$?

Since log is monotonic, $\hat\lambda = \arg\max_\lambda L(\lambda) = \arg\max_\lambda \ell(\lambda)$. Starting from (b), drop $\log(n_j!)$ (no λ-dependence) and split the sum term-by-term, pulling constants out of each:

$$\begin{aligned}
\hat\lambda &= \arg\max_{\lambda>0} \sum_{j\,:\,n_j>0} \Big[-\lambda + n_j\log(\lambda) - \log(1-e^{-\lambda})\Big] \\
&= \arg\max_{\lambda>0} \left\{ \sum_{j\,:\,n_j>0} (-\lambda) + \log(\lambda) \sum_{j\,:\,n_j>0} n_j + \sum_{j\,:\,n_j>0} \big(-\log(1-e^{-\lambda})\big) \right\} \\
&= \arg\max_{\lambda>0} \Big\{ -U\lambda + N\log(\lambda) - U\log(1-e^{-\lambda}) \Big\} \qquad \text{(with } U = \#\{j : n_j > 0\},\ N = \sum_j n_j\text{)}.
\end{aligned}$$

Differentiating w.r.t. λ (chain rule on the last term):

$$\begin{aligned}
\frac{d}{d\lambda}\Big[-U\lambda + N\log(\lambda) - U\log(1-e^{-\lambda})\Big] &= -U + \frac{N}{\lambda} - U\cdot\frac{1}{1-e^{-\lambda}}\cdot\frac{d}{d\lambda}\big(1-e^{-\lambda}\big) \\
&= -U + \frac{N}{\lambda} - U\cdot\frac{e^{-\lambda}}{1-e^{-\lambda}}.
\end{aligned}$$

Set this to zero at $\lambda = \lambda^*$, then gather the U terms before isolating λ^*:

$$\begin{aligned}
-U + \frac{N}{\lambda^*} - U\cdot\frac{e^{-\lambda^*}}{1-e^{-\lambda^*}} &= 0 \\
\frac{N}{\lambda^*} &= U + U\cdot\frac{e^{-\lambda^*}}{1-e^{-\lambda^*}} && \text{(move both } U\text{-terms to the right)} \\
&= U\left(1 + \frac{e^{-\lambda^*}}{1-e^{-\lambda^*}}\right) && \text{(factor } U\text{)} \\
&= U\cdot\frac{1-e^{-\lambda^*}+e^{-\lambda^*}}{1-e^{-\lambda^*}} && \text{(common denominator)} \\
&= \frac{U}{1-e^{-\lambda^*}} && \text{(numerator = 1)} \\
\Longrightarrow\quad \boxed{\frac{\lambda^*}{1-e^{-\lambda^*}} = \frac{N}{U}} && \text{(multiply both sides by } \lambda^*/U\text{)}.
\end{aligned}$$

This is identical to the equation $U/N = (1-e^{-\lambda})/\lambda$ from the Setup: the MLE and the Setup procedure coincide here. The equation has no closed form, so solve numerically by finding the root of

$$g(\lambda) \;=\; \frac{\lambda}{1-e^{-\lambda}} - \frac{N}{U},$$

e.g. by Newton's method, bisection, or fixed-point iteration.

2.4 (d) Estimating M given $\hat{\lambda}$

Question (Ex2.pdf). Once you estimate $\hat{\lambda}$, what is the estimate of N? (Ex2's N = our M, the library complexity.)

The MLE equation in (c) gives $\hat{\lambda}/(1-e^{-\hat{\lambda}}) = N/U$, which alone only reproduces the *observed* count U, not the true library complexity M. To estimate M, applying the Setup substitution $U \approx \mathbb{E}[U] = M\big(1-e^{-\hat{\lambda}}\big)$ gives

$$\hat{M} \;\approx\; \frac{U}{1-e^{-\hat{\lambda}}} \;=\; \frac{N}{\hat{\lambda}},$$

where the second equality is the same rearrangement of the MLE equation as in the Setup. This matches the Setup procedure exactly.

3 Question 2 – Estimating λ on the supplied data

Files. The four supplied files are `Sample1_pt1`, `Sample1_pt2`, `Sample1_pt3`, and `Sample2`. Each row of a file is a tuple $(\text{chr}, \text{start}, \text{end}, n_j)$ with $n_j \geq 1$. From each file we compute $U = \#\{\text{rows}\}$ and $N = \sum_j n_j$.

What "procedure discussed in class" means here. The exercise's single-equation estimator $U/N = (1-e^{-\lambda})/\lambda$ is algebraically the same as the equation we get from $\partial\ell/\partial\lambda = 0$ in Q1(c) (cross-multiply), so a direct comparison is vacuous. The substantive comparison must be against the lecture's library-complexity estimator – a subsampled curve fit of $\mathbb{E}[U] = M(1-e^{-N/M})$ – which is a different procedure and gives a generally different $\hat{M}$. I take that as the intended comparison.

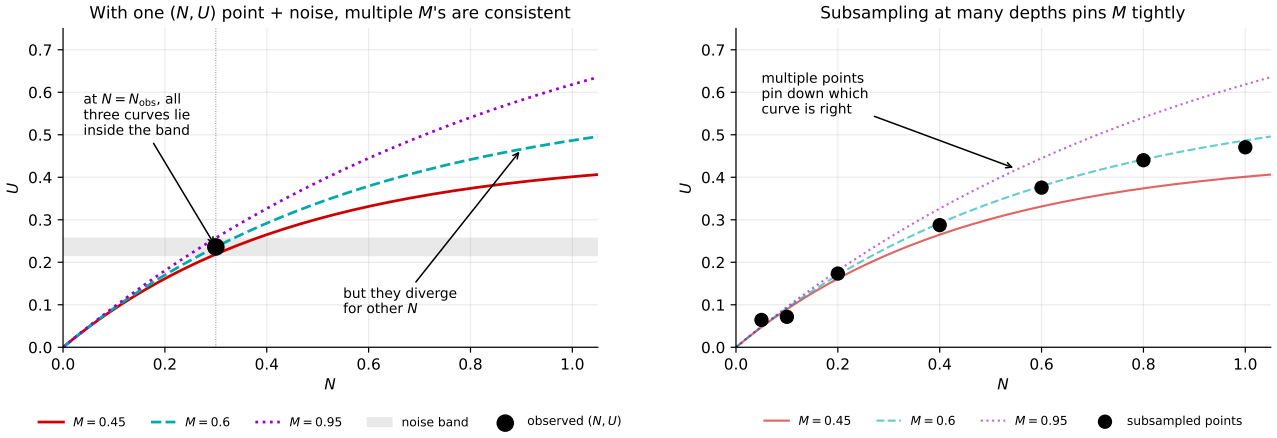


Figure 2: Why the subsampling approach is more informative than the single-point estimator. **Left:** a single observed (N, U) point with a noise band; three different values of M all give predictions inside the band at $N = N_{\text{obs}}$ – the single point cannot tell us which curve generated the data. **Right:** multiple subsampled $(N_{\text{sub}}, U_{\text{obs}})$ points pick out the correct curve unambiguously.

3.1 (a) Q1 MLE vs. subsampling curve-fit on each file

Question (Ex2.pdf). Estimate $\hat{\lambda}$ using both the procedure discussed in class and the maximum likelihood method derived in Q1, on each file. How do the two estimates differ?

Both estimators are run on each file: the Q1 MLE (single-point root-find) and the subsampling curve fit (multi-point least squares).

Method 1: Q1 MLE (single-point root-find)

For each file, $\hat{\lambda}$ solves

$$g(\lambda) \;=\; \frac{\lambda}{1-e^{-\lambda}} - \frac{N}{U} \;=\; 0,$$

and the library-complexity estimate is $\hat{M}_{\mathrm{MLE}} = N/\hat{\lambda}$. The function g is continuous and strictly increasing on $(0,\infty)$, with $g(0^+) = 1 - N/U \le 0$ (since $N \ge U$) and $g(\lambda) \to +\infty$ as $\lambda \to \infty$, so it has a unique positive root.

Numerical method. I use `scipy.optimize.brentq`, a bracketed 1D root finder (a hybrid of bisection and inverse-quadratic interpolation; both bisection and the IVT it relies on were covered in Infi 1). The bracket is $[10^{-10}, N/U]$: at the lower end $g(0^+) = 1 - N/U \le 0$ (since $N \ge U$), and at the upper end $g(N/U) = (N/U)\,e^{-N/U}/(1-e^{-N/U}) > 0$ unconditionally, so the IVT guarantees a root in between. Newton would have worked too but needs a derivative and a good starting point; brentq doesn't.

Method 2: Subsampling curve fit (multi-point least squares)

For each file I generate $(N_{\mathrm{sub}}, U_{\mathrm{obs}})$ pairs at depth fractions $d \in \{0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 1.0\}$, then fit M to the saturation curve through those points.

Producing the $(N_{\mathrm{sub}}, U_{\mathrm{obs}})$ points: Monte Carlo subsampling. For each depth fraction d, randomly pick $m = \lfloor dN \rfloor$ reads without replacement from the bag of N reads, then count how many distinct fragments are present. Each trial is one independent random draw; averaging over 5 seeds reduces sampling noise like $\mathcal{O}(1/\sqrt{\text{trials}})$ by the standard-error-of-the-mean argument.[1]

Efficient implementation. Instead of materialising a flat N-element ticket array (wasteful at $N \sim 4 \times 10^6$), I draw per-fragment counts independently: $\text{count}_j \sim \text{Hypergeometric}(N, n_j, m)$. The marginals are exact, the joint is slightly relaxed (independent draws do not enforce $\sum_j \text{count}_j = m$), but $\mathbb{E}[U_{\mathrm{obs}}] = \sum_j \Pr(\text{count}_j > 0)$ depends only on the marginals, so the estimator is unbiased.

Bonus: closed-form Heck rarefaction.[2] The same expectation has a closed form ($\Pr(\text{miss } j) = \binom{N-n_j}{s}/\binom{N}{s}$; sum the complement):

$$\mathbb{E}[U_{\mathrm{obs}}(s)] \;=\; \sum_{j=1}^{U} \left[\, 1 - \frac{\binom{N-n_j}{s}}{\binom{N}{s}} \,\right], \tag{1}$$

computed in log-space via `scipy.special.gammaln`. I plot it alongside MC in Figure 3 as an independent check.

[1] For i.i.d. trials X_i with finite variance σ^2, $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$, so the s.e. of the sample mean shrinks as $1/\sqrt{n}$ (CLT; see e.g. Casella & Berger, *Statistical Inference*, §5.2).

[2] Full derivation of the subsampling/Heck framework and pedagogical figures in the supplementary *subsampling primer* (`subsampling_primer.pdf`).

Fitting M to the curve. Given the $K = 7$ points $(N_{\text{sub}}^{(i)}, U_{\text{obs}}^{(i)})$, I minimise the sum of squared residuals

$$\hat{M}_{\text{fit}} \;=\; \arg\min_{M>0} \sum_{i=1}^{K} \Big(U_{\text{obs}}^{(i)} - M\big(1 - e^{-N_{\text{sub}}^{(i)}/M}\big) \Big)^2 .$$

This is a one-parameter bounded optimisation. I use `scipy.optimize.minimize_scalar` with bounded Brent's method (the 1D minimisation analogue of `brentq`), bracketing M between $\max_d U_{\text{obs}}(dN)$ (the saturation asymptote forces $M \geq \max_d U_{\text{obs}}$) and a generous ceiling.

Numerical results

The following table combines the Q1 MLE / single-point estimate with the two curve-fit implementations (Heck rarefaction, exact in expectation; and Monte Carlo, averaged over 5 random seeds). The last column reports the inflation ratio $\hat{M}_{\text{MLE}}/\hat{M}_{\text{fit}}$.

File	N	U	$\hat{\lambda}$	$\hat{M}_{\textbf{MLE}}$	$\hat{M}_{\textbf{fit (Heck)}}$	$\hat{M}_{\textbf{fit (MC)}}$	**ratio**
Sample1_pt1	1,666,709	1,176,492	0.7423	2,245,206	2,218,990	2,218,640	1.0118
Sample1_pt2	1,673,083	1,133,365	0.8370	1,998,911	1,975,654	1,975,235	1.0118
Sample1_pt3	1,094,261	759,901	0.7797	1,403,439	1,389,006	1,389,095	1.0104
Sample2	4,006,586	2,332,467	1.2008	3,336,686	3,292,603	3,292,292	1.0134

The saturation curves themselves, one per file, are in Figure 3.

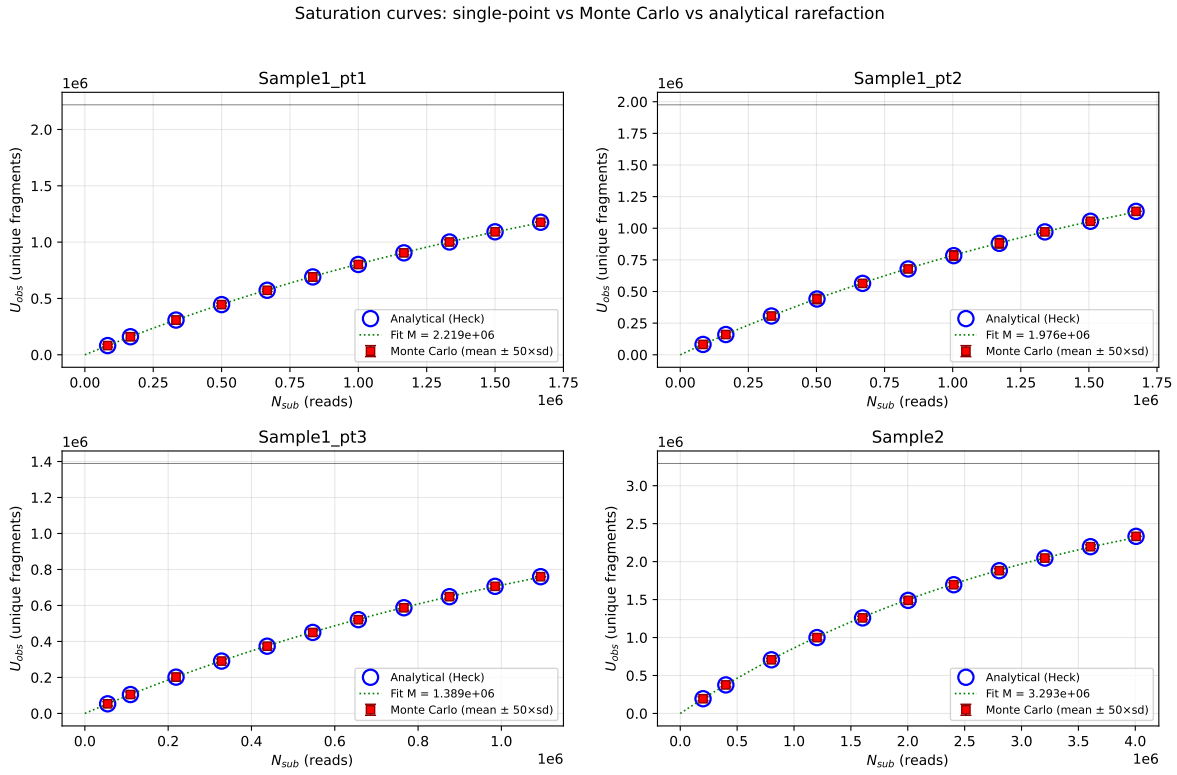


Figure 3: Saturation curves for each file. Analytical rarefaction (blue circles, exact), Monte Carlo (red squares, mean ± s.d. over 5 trials), and the single-point fit (green dotted) are plotted against $(N_{\text{sub}}, U_{\text{obs}})$. The thin horizontal line is the fitted asymptote $\hat{M}_{\text{fit}}$.

Discussion

(i) Monte Carlo agrees with the bonus Heck calculation. On every file the MC- and Heck-based $\hat{M}_{\text{fit}}$ agree to within a few hundred out of millions; Heck (1) is the exact expectation of the MC

procedure, and the residual gap is $\mathcal{O}(1/\sqrt{\text{trials}})$ sampling noise. Both implementations are mutually validated.

(ii) MLE vs subsampling fit: agreement with a tell-tale direction. $\hat{M}_{\text{MLE}}$ and $\hat{M}_{\text{fit}}$ agree within 1.0–1.3% on every file – a small surprise that the whole subsampling+Heck+MC+minimiser pipeline essentially reproduces the one-line single-point root-find. The elaborate fit pays off only as a diagnostic: the consistent *direction* $\hat{M}_{\text{MLE}} > \hat{M}_{\text{fit}}$ across four independent files is too systematic for numerical noise. It signals a model mismatch – the endpoint (N, U) sits below where a true uniform-λ Poisson curve would put it, forcing the single-point estimator to inflate $\hat{M}$. Q2(b) tests this; Q3 finds the mechanism.

(iii) $\hat{M}$ varies across the three Sample 1 parts. $\hat{M}_{\text{fit}} \approx 2.22\text{M}, 1.98\text{M}, 1.39\text{M}$ – the largest is $\sim 1.6\times$ the smallest. Library complexity should be a property of the library, not the depth, so either the three parts are not true subsamples of one library or the saturation model misses something material. Sample 2 sits separately with higher $\hat{\lambda}$ (≈ 1.20 vs. 0.74–0.84) and larger $\hat{M}$, consistent with deeper sequencing.

3.2 (b) Histogram of n_j vs. model prediction; goodness-of-fit

Question (Ex2.pdf). Plot the histogram of n_j in each file vs the model prediction. Do these make sense? Discuss a statistical procedure to judge whether the observation is surprising under the estimate.

Q2(a) surfaced two pieces of indirect evidence that the uniform-λ truncated-Poisson model is misspecified: the systematic 1–1.3% inflation $\hat{M}_{\text{MLE}} > \hat{M}_{\text{fit}}$ on every file, and the $\sim 60\%$ disagreement on $\hat{M}_{\text{fit}}$ across the three Sample 1 parts. Both point toward λ varying across fragments rather than being a single global rate – indeed, fitting a single global λ is operationally an assumption that every fragment shares the same per-molecule sequencing rate, which Q3 will show is false (the $\hat{\lambda}_\ell$ vary by a factor of $\sim$ 5–10 with length). Q2(b) tests that assumption directly: does the empirical histogram of per-fragment read counts match what the truncated-Poisson at $\hat{\lambda}$ predicts?

Setup of the test. For each file, let $\text{obs}[k] = \#\{j : n_j = k\}$ be the number of observed fragments with exactly k reads, for $k = 1, 2, \ldots, n_{\max}$. Under the Q1 model, the conditional distribution of n_j given $n_j \geq 1$ is the zero-truncated Poisson

$$P_{\text{obs}}(n_j = k\,;\,\lambda) \;=\; \frac{e^{-\lambda}\,\lambda^k/k!}{1 - e^{-\lambda}}, \qquad k = 1, 2, 3, \ldots$$

Multiplying by U gives the expected histogram $\exp[k] = U \cdot P_{\text{obs}}(k\,;\,\hat{\lambda})$, i.e. the number of the U observed fragments that should fall in bucket k under the model with $\lambda = \hat{\lambda}$.

Statistical procedure: Pearson's chi-squared goodness-of-fit.[3] The right tool for comparing an observed histogram with a parametric model prediction is the chi-squared goodness-of-fit test. The test statistic is

$$\chi^2 \;=\; \sum_k \frac{\big(\text{obs}[k] - \exp[k]\big)^2}{\exp[k]}.$$

Under the null hypothesis that the truncated-Poisson model is correct, this statistic follows approximately the χ^2 distribution with df = (number of bins) $- 1 - p$ degrees of freedom: the -1 accounts for the total constraint $\sum \text{obs}[k] = U$ and the $-p$ for parameters estimated from the data. Here $p = 1$ because λ was estimated by MLE from the same data (the standard Wilks correction[4]).

[3] Full derivation – variance/CLT chain, $K - 1 - p$ dof from constraints + Wilks, Cochran's rule from Poisson-skewness anchors – in the supplementary *statistics primer* (`stats_primer.pdf`).

[4] Not covered in class. See Wilks, *Ann. Math. Statist.* 9 (1938) 60–62 for the original asymptotic-χ^2 result; also Casella & Berger, *Statistical Inference*, §10.3.1.

Cochran tail-merging.[5] The histogram bins here are the per-fragment read-count buckets $k = 1, 2, 3, \ldots$, not the 5-bp length bins of Q3. The χ^2 approximation requires every bin to have $\exp[k] \geq 5$, otherwise the discrete sum behaves badly under the continuous χ^2. Sparse right-tail bins (high k, where the truncated-Poisson decays exponentially) routinely fall below this threshold, so I merge from the largest k downwards into the next bin, accumulating both observed and expected counts, until every retained bin satisfies $\exp[k] \geq 5$.

Decision rule. The p-value is $p\text{-value} = 1 - F_{\chi^2}(\chi^2_{\text{obs}}; \text{df})$. A small p-value (say < 0.05) means the observed histogram is unlikely under the model, i.e. the uniform-λ truncated Poisson is rejected for that file. A large p-value means the model is consistent with the data at the tested level (it does *not* confirm the model – absence of evidence is not evidence of absence – but it places no constraint on rejection).

Numerical results. I run the test with both estimates of λ from Q2(a): the single-point MLE $\hat{\lambda}_{\text{sp}}$ and the subsampling curve fit $\hat{\lambda}_{\text{curve}}$ (the latter is slightly larger, consistent with the smaller $\hat{M}_{\text{fit}}$ from Q2(a) – $\lambda = N/M$, so smaller M gives larger λ). Per-file χ^2 values:

File	$\hat{\lambda}_{\text{sp}}$	$\hat{\lambda}_{\text{curve}}$	χ^2_{sp}	χ^2_{curve}
Sample1_pt1	0.742	0.751	23,911	23,111
Sample1_pt2	0.837	0.847	24,529	23,659
Sample1_pt3	0.780	0.788	11,402	11,094
Sample2	1.201	1.217	54,731	53,282

After Cochran tail-merging on the per-fragment read-count histogram, every file retains roughly 5–7 retained bins (collapsing the long sparse right-tail into a single "tail" bin), giving df $\approx$ 4–6. This is not destructive: most of the data sit in the first few low-k bins where the model already fails egregiously; the tail merge only removes bins with too little data to test individually. The critical χ^2_5 at $\alpha = 0.05$ is ≈ 11.07; at $\alpha = 10^{-6}$ it is ≈ 32. The observed values are in the *tens of thousands*, so $p \approx 0$ at any sensible threshold. The truncated-Poisson model is rejected on every file regardless of the dof correction.

Figure 4 shows the four histograms with both prediction curves overlaid.

Discussion. (i) The two $\hat{\lambda}$ values give essentially the same χ^2. $\hat{\lambda}_{\text{sp}}$ and $\hat{\lambda}_{\text{curve}}$ are within $\sim 1\%$ (the same agreement as $\hat{M}$ from Q2(a) by $\hat{\lambda} = N/\hat{M}$), so the two χ^2 columns agree to a few percent. At this scale of misfit ($\chi^2 \sim 10^4$) the difference is negligible.

(ii) The model has to be rejected outright. χ^2 in the tens of thousands with df ≈ 5 is not borderline. The failure is visible in Figure 4: observed bars overshoot the predicted curve by orders of magnitude in the right tail. The Q1/Q2(a) chain – truncated-Poisson likelihood, MLE for λ, saturation curve $\mathbb{E}[U] = M(1 - e^{-N/M})$ – rests on a foundation the data refute, so $\hat{\lambda}$ and $\hat{M}$ should be read as descriptive summaries, not "true" Poisson rate or library complexity.

(iii) The mechanism is heterogeneous λ, bridging to Q3. A heterogeneous-λ Poisson model – each fragment has its own rate λ_j – can produce arbitrarily heavy right tails: high-λ fragments populate the over-represented $n_j \approx$ 8–12 bins while low-λ ones dominate at $n_j = 1, 2$. This is also what Q2(a)'s $\hat{M}_{\text{MLE}} > \hat{M}_{\text{fit}}$ inflation gap pointed at indirectly. Q3 makes the heterogeneity explicit by stratifying on fragment length.

[5] Also not covered in class. Cochran, *Biometrics* 10 (1954) 417–451 – the canonical reference; modern statistics texts (e.g. Agresti, *Categorical Data Analysis*, §3.2) restate it as $\exp[k] \geq 5$ for every bin (some allow ≥ 1 for a few bins).

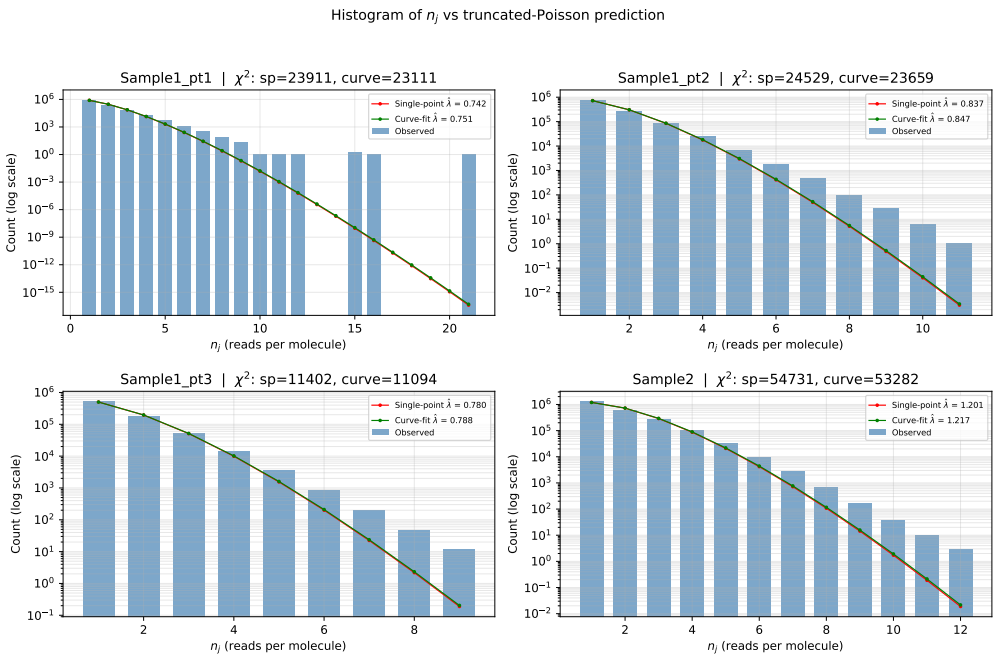


Figure 4: Observed multiplicity histograms (blue bars, log y-axis) vs. the zero-truncated Poisson prediction at $\hat{\lambda}_{\text{sp}}$ (red) and $\hat{\lambda}_{\text{curve}}$ (green). The two prediction curves lie essentially on top of each other because the $\hat{\lambda}$ values differ by $< 1\%$. The bars sit visibly *above* the curves throughout the right tail on every file, the signature of heavier-than-Poisson per-fragment counts.

4 Question 3 – Fragment-length dependence of PCR duplication

PCR efficiency at length ℓ. The per-cycle probability that a template of length ℓ is successfully copied during a PCR step (equivalently, the multiplicative growth factor for templates of that length per cycle). It is set by the protocol – enzyme, primers, cycle number, denaturation/annealing timing – and depends on length because longer templates take more time per extension and are more prone to mid-extension stalling, polymerase fall-off, and strand breakage.

Predicted shape: PCR length bias. PCR length bias (discussed in class) favours short templates, so pure PCR efficiency would give a monotonically decreasing $\lambda(\ell)$. The observed curves are not monotone: a sharp left-edge rise (upstream size selection, no input below $\sim$ 100–150 bp) then a slow right-side decay. Only the right-side decay matches the pure-PCR prediction; the monotone prediction holds only *above* the size-selection cutoff.

What λ_ℓ actually measures. Per-bin λ_ℓ is not a pure measurement of PCR efficiency. It conflates three contributions:

1. the per-cycle PCR amplification efficiency at length ℓ,
2. the abundance of length-ℓ molecules in the pre-PCR library, and
3. how the sequencer's reads were distributed across fragments (mappability, flowcell loading, etc.).

Attributing the observed $\lambda(\ell)$ to (1) alone requires controlling for (2) and (3); Q3(d) takes this up.

4.1 (a) λ vs. fragment length

Question (Ex2.pdf). Bin the reads by length (5 bp increments), estimate λ in each bin, and plot λ vs fragment length.

Implementation. For each file I bin fragments in 5 bp increments, compute (N_ℓ, U_ℓ) per bin, and estimate $\hat{\lambda}_\ell$ by the single-point root-find via `brentq` (Q2(a) Method 1). Bins with $U_\ell < 50$ are dropped as too noisy. I use single-point rather than the subsampling curve fit because (i) per-bin sample sizes are too small for the saturation curve to be well-constrained, and (ii) the LRT in (b) needs MLEs (Wilks' theorem), which single-point provides directly.

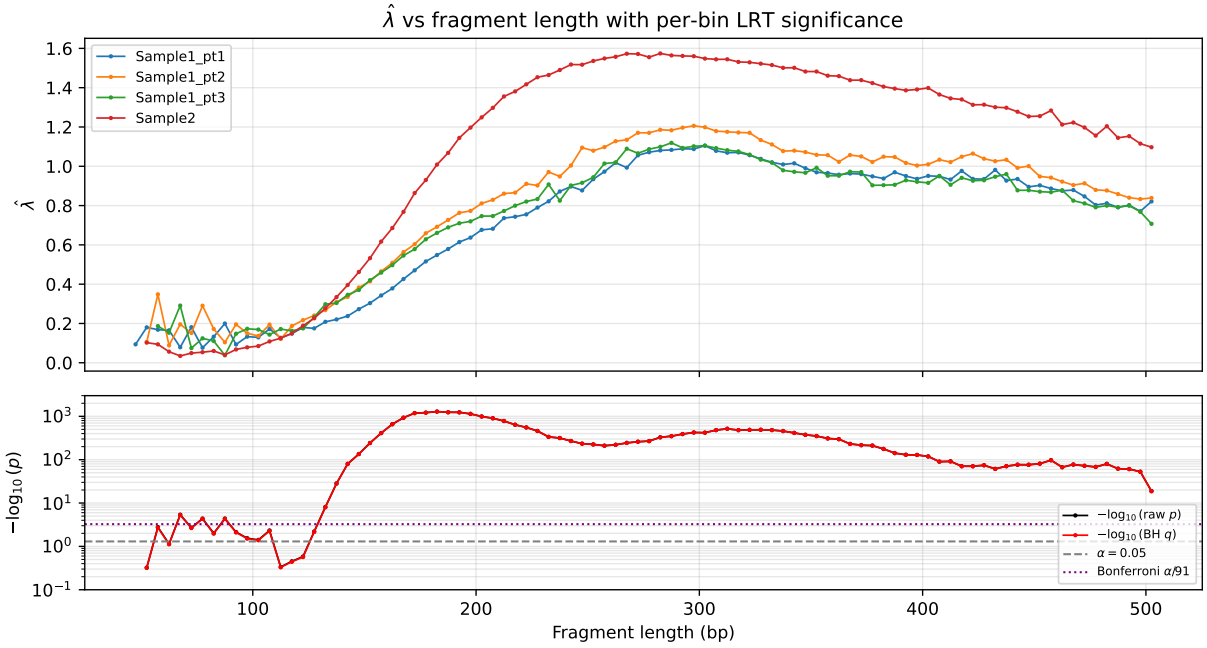


Figure 5: **Top:** $\hat{\lambda}_\ell$ vs fragment length, one curve per file. **Bottom:** per-bin LRT $-\log_{10}(p)$ (raw p black, BH q red), log y-axis; **higher = stronger rejection of** H_0. Reference lines: $\alpha = 0.05$ (grey dashed), Bonferroni $\alpha/91$ (purple dotted) – reject above either line.

Per-file summary.

File	peak $\hat{\lambda}_\ell$	peak at (bp)	mean $\hat{\lambda}_\ell$	retained bins
Sample1_pt1	1.105	302	0.707	92
Sample1_pt2	1.206	298	0.801	91
Sample1_pt3	1.119	288	0.738	90
Sample2	1.574	282	1.069	91

The retained length range is $\sim[48, 502]$ bp across all files (bins with $U_\ell < 50$ discarded). All four peak between 282 and 302 bp – a 20 bp window – while peak heights span a much wider range (1.10–1.57). Mean $\hat{\lambda}_\ell$ averaged across all bins ranks the files in the same order as the global $\hat{\lambda}$ from Q2(a), as expected.

4.2 (b) Comparing $\lambda(\ell)$ across files; significance

Question (Ex2.pdf). How do these functions differ between the different sequencing files? How would you assess if they are significantly different?

Patterns visible by eye in Figure 5. (P1) Sample 2 sits uniformly 40–50% **above the Sample 1 parts above the size-selection cutoff ($\ell \gtrsim 120$ bp).** Shape (peak, asymmetry, decay) is identical; only the amplitude is shifted. Consistent with deeper per-molecule sequencing at a shared protocol – Q2(a)'s global $\hat{\lambda}$ ratio $1.20/0.78 \approx 1.54$ matches the per-length amplitude shift.

(P2) The three Sample 1 parts trace nearly the same curve (within $\sim$ 10% in amplitude).

A meaningful improvement over Q2(a), where the global $\hat{M}_{\text{fit}}$ values disagreed by $\sim 60\%$ – evidence that length is the dominant heterogeneity axis Q2(b) detected.

(P3) The peak location is shared across all four files ($\ell \approx 290\,\text{bp} \pm 10$). Set by library-prep target insert size and the PCR sweet spot, both shared across the same protocol (discussed in Q3(c)).

(P4) Sample 2 is visibly smoother because each bin holds $\sim 2\times$ more fragments and the per-bin variance scales as $1/U_\ell$ – noise, not a different process.

Formal significance test: the likelihood-ratio test (LRT). Compare H_0 (all $F = 4$ files share one rate λ_ℓ at this bin) against H_1 (each file has its own $\lambda_\ell^{(f)}$, $f = 1, \ldots, F$). H_0 is nested in H_1, so H_1 never fits worse; the question is whether it fits *much* better (real differences) or only marginally (sampling noise). *Full derivation in the supplementary statistics primer (`stats_primer.pdf`).*

The statistic. For each file f, the truncated-Poisson log-likelihood derived in Q1(b) is

$$\ell_f(\lambda) \;=\; N_\ell^{(f)} \log \lambda \;-\; U_\ell^{(f)} \lambda \;-\; U_\ell^{(f)} \log\!\big(1 - e^{-\lambda}\big) \;+\; \text{const.}$$

The classical *Neyman–Pearson likelihood ratio* for two simple hypotheses is $\Lambda_{NP} = L(\hat{\theta}_{H_0})/L(\hat{\theta}_{H_1})$ with range $(0, 1]$, since $L(\hat{\theta}_{H_1}) \geq L(\hat{\theta}_{H_0})$ by nesting. We use the inverted, twice-logged, multi-file form (standard for Wilks):

$$\boxed{\Lambda_\ell \;=\; 2\Big[\textstyle\sum_f \ell_f(\hat{\lambda}_\ell^{(f)}) \;-\; \sum_f \ell_f(\hat{\lambda}_\ell^{(0)})\Big]}$$

where $\hat{\lambda}_\ell^{(f)}$ is each file's own MLE and $\hat{\lambda}_\ell^{(0)}$ is the pooled MLE. By construction $\Lambda_\ell \in [0, \infty)$ – per file, how much better fitting its own MLE does than being forced onto the shared one, summed; *no upper bound*, unlike Λ_{NP}.

By Wilks' theorem (1938 – not covered in class), under H_0 and mild regularity, Λ_ℓ is asymptotically χ^2-distributed with df $= F - 1 = 3$ (the parameter-count difference, $p_1 - p_0 = F - 1$). Thus $p_\ell = \Pr\big[\chi^2(F-1) > \Lambda_\ell\big]$.

Multiple-testing correction. With 91 simultaneous tests I apply both corrections discussed in Lecture 5: *Bonferroni* (controls FWER; threshold $\alpha/91 \approx 5.5 \times 10^{-4}$; safe but conservative) and *Benjamini–Hochberg* (controls FDR; adaptive step-up procedure rejecting at $p_{(k)} \leq (k/91)q$). Reporting both lets the conclusion be checked for robustness; BH is the more powerful here since Bonferroni already shows most nulls are false.

Result. Running the test on the four files at the 91 length bins gives:

Rejection criterion	**Bins rejecting H_0**	**Fraction**
Raw $p < 0.05$ (no correction)	86	95%
Benjamini–Hochberg $q < 0.05$ (FDR)	86	95%
Bonferroni $p < 5.5 \times 10^{-4}$ (FWER)	78	86%

The curves are genuinely different across files, robustly to the choice of correction. The raw-p and BH-q counts coincide because the LRT p-values are sharply bimodal: each bin's p is either far below any threshold (true rejection, mostly driven by Sample 2 vs Sample 1) or comfortably above 0.05 (no signal). The Bonferroni count is slightly lower because its threshold ($\alpha/91$) is about $90\times$ tighter than 0.05, so a handful of mid-magnitude p-values that BH/raw retain are pushed back into "not rejected". The same overall picture nonetheless holds under every correction: at most lengths the four files have detectably different rates.

Why $\sim 95\%$ rejection when the Sample 1 parts are supposed to be replicates? Two reasons: **(a)** the LRT is jointly on all four files, and the Sample 1 vs Sample 2 amplitude gap (P1) dominates most bins, so most rejections correctly flag Sample 2; **(b)** with per-bin U_ℓ in the tens of thousands, even

a 10% within-Sample-1 spread is statistically detectable – the test correctly says the four $\hat{\lambda}_\ell$ are not *exactly* equal, not that the differences are biologically meaningful. Pairwise S1-only LRTs (not run here) would separate the S1-vs-S2 signal from S1-internal noise.

4.3 (c) What the curves say about PCR + sequencing bias

Question (Ex2.pdf). What do these graphs tell you about biases in the PCR + sequencing process?

The four curves agree on a striking qualitative shape – a steep rise, a peak near $\ell \approx 290\,\text{bp}$, and a long slow tail. Each feature has a biological interpretation.

The shape can be read as three regimes, each with its own mechanism.

Short side ($\ell \lesssim 120\,\text{bp}$): size-selection. Library prep includes a SPRI / gel size-selection step that physically removes fragments below $\sim$ 100–150 bp before sequencing. $\hat{\lambda}_\ell$ is small and noisy there because the input is absent, not because PCR failed.

Peak ($\ell \approx 290\,\text{bp}$). Two reinforcing effects: protocols target a 200–400 bp insert range so the pre-PCR pool is densest there, and PCR efficiency is highest in the same range (reliable annealing, fast extension, low fall-off). All four files peaking at the same length supports a common protocol target.

Long side: PCR length bias. PCR efficiency declines gradually with length (polymerase fall-off, strand breaks, harder strand-displacement), geometrically per cycle; after $\sim$ 10–15 cycles this accumulates into the slow decay seen here. The asymmetric shape (steep left, gradual right) is the fingerprint of a physical cutoff on the short side and a continuous efficiency decline on the long side.

Connection back to Q2(b). $\hat{\lambda}_\ell$ varies by a factor of $\sim$ 5–10 across length; the global $\hat{\lambda}$ from Q1/Q2 averages over this variation, which is why the global truncated-Poisson fits so badly. Stratifying by length is the natural fix for this axis of bias.

4.4 (d) Reconstructing the original fragment-length distribution

Question (Ex2.pdf). Can you reconstruct the presumed fragment-length distribution in the original library? Explain how you go about it and what you learned.

Within-bin homogeneity assumption. The model below assumes a single λ_ℓ governs every fragment in length bin ℓ. Q3(b) gives indirect support – the three Sample 1 replicates agree within $\sim 10\%$ at every bin, consistent with one common per-bin rate – but this is not a goodness-of-fit test. A formal check (per-bin Pearson χ^2, as in Q2(b)) is not asked for and was not run, so the $\hat{M}_\ell$ below inherit whatever within-bin heterogeneity persists.

Method. Per length bin ℓ, the truncated-Poisson model says

$$\mathbb{E}[U_\ell] \;=\; M_\ell\big(1 - e^{-\lambda_\ell}\big),$$

where M_ℓ is the (unknown) pre-PCR molecule count at length ℓ. Substituting the observed U_ℓ for $\mathbb{E}[U_\ell]$ and $\hat{\lambda}_\ell$ from Q3(a),

$$\hat{M}_\ell \;\approx\; \frac{U_\ell}{1 - e^{-\hat{\lambda}_\ell}} \;=\; \frac{N_\ell}{\hat{\lambda}_\ell}$$

(the second equality holds at the MLE, by the same algebra as Q1(d) and the Setup of Q2).

Why dividing by $(1 - e^{-\hat{\lambda}_\ell})$ undoes the length bias. Q3(a)–(c) established that $\hat{\lambda}_\ell$ is not flat: it peaks near 290 bp – PCR's sweet spot – and falls off in both directions. Consequently the detection probability $(1 - e^{-\hat{\lambda}_\ell})$ is also length-dependent: high (close to 1) at lengths PCR amplifies well, lower (down to near 0.4) at lengths it does not. Because $\mathbb{E}[U_\ell] = M_\ell(1 - e^{-\lambda_\ell})$, the un-normalised observed

count U_ℓ already mixes the true abundance M_ℓ with the length-dependent detection probability – so a raw histogram of U_ℓ is biased toward lengths PCR favours. Dividing by the *same length-specific* detection probability removes that mixing factor: $\mathbb{E}[U_\ell/(1-e^{-\lambda_\ell})] = M_\ell$, i.e. the estimator is *unbiased* for the pre-PCR count M_ℓ in the statistical sense that its expected value equals the parameter. Bins PCR over-amplified get a near-1 correction; bins it neglected get a large boost that recovers (in expectation) the silent molecules that produced zero reads.

Is this enough? It is sufficient to invert the length axis of the bias. It is *not* a guarantee that all PCR-induced distortion vanishes: any heterogeneity orthogonal to length – GC content, sequence context, mappability, position effects – is not corrected by this inversion and remains baked into $\hat{M}_\ell$. The estimator returns the pre-PCR length distribution that is consistent with treating length as the dominant bias axis, not absolute molecule counts. See *Sufficiency check* below for an empirical test of how much the correction actually achieves on these data.

Normalising. The normalised sequence $\hat{p}_\ell^{(\mathrm{pre})} = \hat{M}_\ell / \sum_{\ell'} \hat{M}_{\ell'}$ is a probability distribution – the fraction of the pre-PCR library at each length, with the library-size scale divided out so files of different total sizes are comparable. The raw observed $\hat{p}_\ell^{(\mathrm{obs})} = U_\ell / \sum U_{\ell'}$ is computed alongside as the post-PCR baseline.

Implementation. `q3d.py` extends `q3a.py` with the per-bin $\hat{M}_\ell$ and the normalised distributions $\hat{p}_\ell^{(\mathrm{pre})}$ and $\hat{p}_\ell^{(\mathrm{obs})}$, computes per-distribution summary statistics, and runs the sufficiency check below.

Results.

	Observed $\hat{p}_\ell^{(\mathbf{obs})}$ (post-PCR)					**Reconstructed $\hat{p}_\ell^{(\mathbf{pre})}$ (pre-PCR)**				
File	peak	mean	med.	std	IQR	peak	mean	med.	std	IQR
Sample1_pt1	178	258	208	97	153	168	232	180	94	144
Sample1_pt2	168	252	196	96	151	168	233	181	94	145
Sample1_pt3	168	253	198	96	152	168	235	182	95	146
Sample2	182	243	224	85	116	**142**	**206**	**181**	92	121

All units bp; lengths are bin midpoints. Two patterns stand out.

(i) The reconstructed distribution sits to the left of the observed one, on every statistic: peak drops by 0–40 bp, mean by 17–37 bp, median by 0–43 bp. This is the PCR-bias correction working as intended – $\hat{\lambda}_\ell$ peaks near $\ell \approx 290$ bp, so fragments in that range are over-represented in U_ℓ; dividing by $(1-e^{-\hat{\lambda}_\ell})$ pulls probability mass back toward the short end where the underlying library has more molecules per read.

(ii) Sample 2's shift is much larger than Sample 1's. Sample 2 has uniformly higher $\hat{\lambda}_\ell$ (Q3(b)), so its absolute correction factor $1/(1-e^{-\hat{\lambda}_\ell})$ is closer to 1 everywhere; but the peak location is set by the *relative* shape of the correction across bins. The relative shape pulls Sample 2 from 182 → 142 bp; the three Sample 1 parts share a different relative shape that leaves the peak at 168 bp. After correction, however, the median converges – 180–182 bp for all four files – which is exactly the cfDNA signature (see biological reading below).

Biological reading: why two peaks, not one. The reconstructed distribution has a dominant peak near 168 bp (Sample 1) or 142 bp (Sample 2) and a secondary shoulder near 320 bp that survives the correction. This is the classic cell-free DNA fragmentation pattern: cfDNA is preferentially cleaved in nucleosome linker regions, leaving fragments at integer multiples of the ~ 167 bp nucleosome footprint – a sharp mono-nucleosome peak and a di-nucleosome shoulder near ~ 330 bp. The presence of *both* peaks after correction indicates they are not PCR artefacts but real features of the input library.

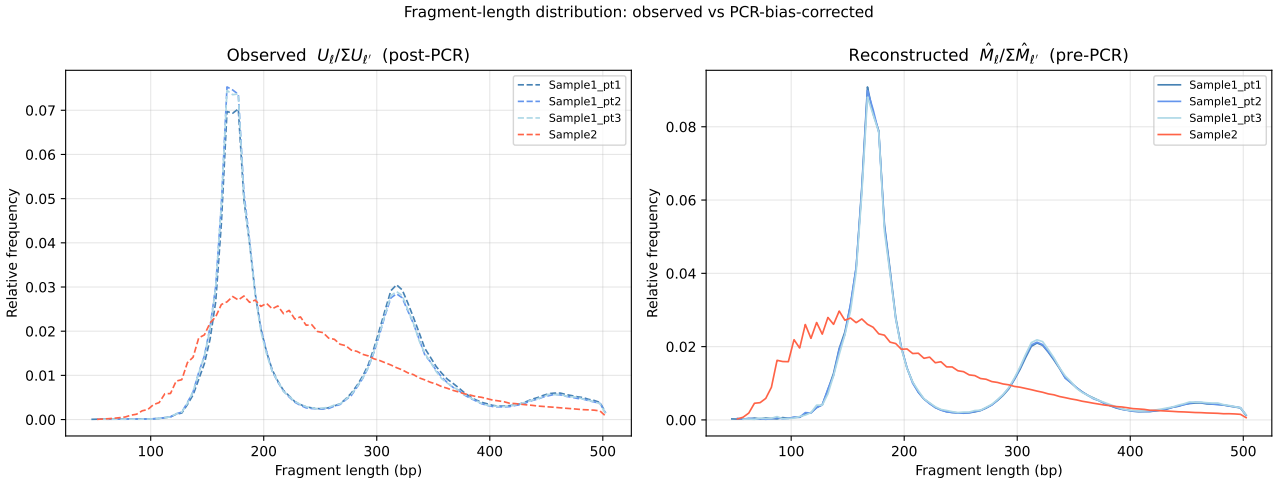


Figure 6: Fragment-length distribution before and after PCR-bias correction. **Left:** raw observed distribution $\hat{p}_\ell^{(\mathrm{obs})} = U_\ell / \sum U_{\ell'}$ – biased toward the PCR sweet spot near $\ell \approx 290$ bp. **Right:** reconstructed pre-PCR distribution $\hat{p}_\ell^{(\mathrm{pre})} = \hat{M}_\ell / \sum \hat{M}_{\ell'}$, dividing through by the per-bin detection probability $(1 - e^{-\hat{\lambda}_\ell})$. The dominant peak near $\ell \approx 168$ bp and the secondary shoulder near $\ell \approx 320$ bp persist in both panels.

Sample 2's sharper, shorter peak (142 vs 168 bp, smaller IQR) is consistent with a more degraded or differently-prepared cfDNA library.

Sufficiency check. A free internal test of whether the correction does anything useful: the three Sample 1 parts are presumed replicates of one library, so their reconstructed distributions should agree more closely than their observed distributions. Using total variation distance (TVD),

Pair	TVD **(obs)**	TVD **(pre)**	**change**
pt1 vs pt2	0.032	0.008	−0.024
pt1 vs pt3	0.028	0.019	−0.009
pt2 vs pt3	0.008	0.015	+0.007

Two of three pairs move closer; the average TVD across the three pairs drops from 0.023 to 0.014 ($\sim 40\%$ reduction). The pt2–pt3 pair was already the closest in observed space, so its slight regression is within the noise of the per-bin $\hat{\lambda}_\ell$ estimates. Net: the correction provides a measurable but partial improvement in replicate consistency, which is what one expects if length is one important but not exhaustive bias axis.

Direction-correct, magnitude unverifiable. The TVD check says $\hat{M}_\ell$ moves the right way, but can't tell under- from over-correction: within-bin heterogeneity could leave residual PCR bias in $\hat{M}_\ell$, and multiplying $1/(1-e^{-\hat{\lambda}_\ell})$ into low-$\hat{\lambda}$ bins driven by size-selection (not PCR inefficiency) could fabricate a left-edge bump. Without a ground-truth pre-PCR distribution the absolute magnitude is unverifiable.

What we learned. (1) The pre-PCR length distribution is recovered by $\hat{M}_\ell = U_\ell/(1 - e^{-\hat{\lambda}_\ell})$. (2) After correction all four files share a median near 180 bp – the mono-nucleosome footprint – with a di-nucleosome shoulder near ~ 320 bp, consistent with a cfDNA library. (3) The correction tightens Sample 1 replicate agreement by $\sim 40\%$ in TVD, so length is real but not the only bias axis. Size selection itself is not invertible – the small U_ℓ at $\ell \lesssim 100$ bp reflects fragments physically removed before sequencing, which no inversion can recover – and Q2(b)'s rejection of the global Poisson means absolute molecule counts would still need other axes (GC, context, position) modelled in.